

Rebuttal

Links

文章链接——不用交新版本

<https://www.overleaf.com/project/6972d1eddec304cecfebe028>

图表

<https://www.figma.com/design/a0P0bfbeDnXcuhKawKWWJ1/Untitled?node-id=154-254&t=eTj28vptk4cysLfa-1>

Rebuttal Guidelines

The goal of the rebuttal/response process is to help ensure that paper submissions receive a fair and accurate assessment:

For early accepted papers: authors have the option to respond to possible misinterpretations or inaccuracies in the reviewers' evaluations (i.e., optional rebuttal). This may help inform the selection of orals and awards. The response will be published alongside the reviewer comments.

For rebuttal papers: authors have the opportunity to highlight possible misinterpretations or inaccuracies in the reviewers' assessment and respond to reviewer comments. This will help inform the reviewers' and Area Chairs' final recommendation. Rebuttals will be published alongside the reviewers' comments if the paper is accepted.

Rebuttals will be text only and limited to 4,000 characters. Responses exceeding 4,000 characters will not be reviewed. Authors are reminded that links to external material are not allowed to be included in the rebuttal. This includes links to code repositories. **In the rebuttal, authors can indicate their intention to make code available after acceptance.** Please note that anonymity should be maintained during the rebuttal process, and breaking it at this stage can result in rejection.

An effective rebuttal focuses on the major issues raised in the reviews. These should be addressed before responding to minor points. By prioritizing and grouping multiple reviewer comments that generally pertain to the same issue into a few major categories, you demonstrate that you understand the primary concerns of reviewers.

It is recommended that you **summarize or paraphrase the issue before you address it**, and clearly indicate to which comment(s) you are responding. While the space for rebuttal is limited, if properly utilized by condensing the response down to the essentials, this is an effective way to show that you understood the reviewers' concerns and have valid answers to the questions

raised in the reviews, or to establish that certain reviewer comments were incorrect or unsubstantiated.

An effective rebuttal addresses reviewers' criticisms by explaining where in the paper you had provided the requisite information, and perhaps further clarifying it.

New/additional experimental results in the rebuttal are not allowed, and breaking this rule is grounds for automatic desk rejection. It is, however, allowed to amend the presentation of existing results. Please note that the final paper decision will be made based on the content of the submitted manuscript and not any promised changes in the rebuttal. If accepted, authors are not allowed to make substantial changes to the fundamental content of the paper, including experiments, data, and analysis.

In the case that a reviewer requests specific, additional experiments or results as part of the rebuttal, Area Chairs are instructed to ignore those requests and not to penalize rebuttals that adhere to the rebuttal guidelines. Authors may instead address such concerns primarily based on the content of the submitted paper. For example, authors may want to acknowledge suggestions for further experiments and consider them for future work. Authors may also decide to refute a request by making a case why a requested experiment is not necessary to judge their presented contributions.

Finally, an effective rebuttal should remain polite and professional. If you believe a review was discourteous or included incorrect or unsubstantiated claims, you can clearly and concisely address them in your rebuttal.

View Reviews

Paper ID

4292

Paper Title

LongMedBench: Evaluating Medical Agents in Long-Horizon Clinical Reasoning and Decision Making

Reviewer #1

Questions

3. Please categorize the relevance of the paper (you may choose more than one). Note the different assessment criteria for the different paper categories as outlined in the Reviewer Guidelines:

<https://conferences.miccai.org/2026/en/REVIEWER-GUIDELINES.html>

- CAI

4. How would you describe the paper?

- Methodological contribution

5. Please describe the main contribution of the paper.

- The paper starts with constructing a dataset architecture and a taxonomy for studying the medical agents' temporal sensitivity. The paper then runs experiments to evaluate medical agents in long-horizon clinical reasoning. The paper finds that LLM agents struggle with implicit temporal reasoning for visit-level understanding for patients.

6. Please list the major strengths of the paper. For example, you could highlight a novel formulation, an original way to use data, a demonstration of clinical feasibility, a novel application, a particularly strong evaluation, or anything else that is a strong aspect of this work. Please provide details, for instance, if a method is novel, explain what aspect is novel and why this is interesting.

- The paper studies an interesting problem. Understanding the medical LLM agents is important.

7. Please list the major weaknesses of the paper. Please provide details: for instance, if you state that a formulation, way of using data, demonstration of clinical feasibility, or application is not novel, then you must provide specific references to prior work.

- The paper is not very well written, there are many things undefined and not well explained. Moreover, the contribution is incremental. The figures are not very clear. Specifically, the example in figure 2 is not well explained and it does not add understanding to the reader. Moreover, the work is not reproducible. I state here some minor comments on improvements, typos and errors:
 - Do not capitalize after the semicolon
 - The bold text is excessive
 - On page 2, paradigms is going out of paper margin
 - Add the article "the" when mentioning the agent
 - Add a space between the word and the references
 - Explain what a medical event is
 - The patient record S, should this have some superscript to indicate which patient?
 - On page 4, "an action" not "a action"
 - On page 5, "No external tools are" not "is"
 - On page 6, "We introduce Mem0" but isn't Mem0 coming from reference [1]? Therefore, you are not introducing it.
 - Also about Mem0, please add an explanation to what it is.
 - The references should be capitalized correctly.

8. Please rate the clarity and organization of the paper.

- Poor

9. Please comment on the reproducibility of the paper. Please be aware that providing code and data is a plus, but not a requirement for acceptance.

- The submission does not provide sufficient information for reproducibility.

10. Code of Ethics Check. Based on your review and your understanding of the MICCAI Scientific Code of Ethics, do you believe this submission may involve a potential ethics concern or violation? Reviewers are expected to familiarize themselves with the MICCAI Scientific Code of Ethics before completing this assessment: <https://miccai.org/index.php/about-miccai/policies/scientific-code-of-ethics/>

- No

13. Rate the paper on a scale of 1-6, 6 being the strongest (6-4: accept; 3-1: reject). Please use the entire range of the distribution. Spreading the score helps create a distribution for decision-making.

- 3. Weak Reject — marginally below the acceptance threshold, but would not mind if accepted, dependent on rebuttal

14. Please justify your recommendation. What were the major factors that led you to your overall score for this paper?

- As stated in the weaknesses section, the paper's contribution is minor and it needs to be better written for easier understanding of the reader.

16. In view of your answers above and your overall experience, how would you rate your confidence in your review?

- Confident but not absolutely certain (3)

Reviewer #2

Questions

3. Please categorize the relevance of the paper (you may choose more than one). Note the different assessment criteria for the different paper categories as outlined in the Reviewer Guidelines:

<https://conferences.miccai.org/2026/en/REVIEWER-GUIDELINES.html>

- MIC
- CAI

- Clinical Translation of Methodology

4. How would you describe the paper?

- Methodological contribution

5. Please describe the main contribution of the paper.

- This paper introduces LongMedBench, a MIMIC-IV-based benchmark for evaluating medical agents on longitudinal clinical reasoning across many visits. It organizes evaluation into three task groups: factual retrieval, temporal reasoning, and long-horizon decision-making. The contribution is relevant, but it should be framed as a more temporally focused benchmark design, rather than a wholly new direction, given prior work such as MedAgentBench (Jiang et al., 2025), AgentClinic (Schmidgall et al., 2024), TIMER (Cui et al., 2025), ER-Reason (Mehandru et al., 2025), and EHRSQL (Lee et al., 2022).

6. Please list the major strengths of the paper. For example, you could highlight a novel formulation, an original way to use data, a demonstration of clinical feasibility, a novel application, a particularly strong evaluation, or anything else that is a strong aspect of this work. Please provide details, for instance, if a method is novel, explain what aspect is novel and why this is interesting.

- The paper targets an important problem: real clinical reasoning is longitudinal, and most current medical LLM evaluations still under-test this setting. The benchmark design is reasonably well structured, especially the split between note memory, event memory, and contextual memory, and the temporal reasoning tasks are the strongest part of the paper. In particular, the benchmark clearly separates explicit timestamp use from implicit temporal inference, and the results show that models perform much better on the former than the latter. That is a useful and believable finding. The negative result is also valuable: memory or retrieval helps factual lookup more than actual downstream decision-making.

7. Please list the major weaknesses of the paper. Please provide details: for instance, if you state that a formulation, way of using data, demonstration of clinical feasibility, or application is not novel, then you must provide specific references to prior work.

- The main weakness is that the paper **overclaims novelty**. Several recent benchmarks already move in this direction, including MedAgentBench (Jiang et al., 2025), AgentClinic (Schmidgall et al., 2024), TIMER (Cui et al., 2025), and ER-Reason (Mehandru et al., 2025). **So the novelty is more in the specific task packaging than in the overall idea.** A second issue is that the “long-horizon decision-making” task may not truly measure optimal clinical reasoning, because **predicting the next recorded EHR action is closer to trajectory continuation than gold-standard decision evaluation.** Third, the paper’s own ablations show that adding more historical memory gives limited benefit, which weakens the claim that the benchmark already isolates a strong long-horizon reasoning bottleneck. ~~Finally, there are some inconsistencies that should be fixed, such as 335 vs. 355 patients and the stated 128K context limit versus 400K for GPT-5-mini in Table 4.~~

8. Please rate the clarity and organization of the paper.

- Good

9. Please comment on the reproducibility of the paper. Please be aware that providing code and data is a plus, but not a requirement for acceptance.

- The submission does not provide sufficient information for reproducibility.

10. Code of Ethics Check. Based on your review and your understanding of the MICCAI Scientific Code of Ethics, do you believe this submission may involve a potential ethics concern or violation? Reviewers are expected to familiarize themselves with the MICCAI Scientific Code of Ethics before completing this assessment: <https://miccai.org/index.php/about-miccai/policies/scientific-code-of-ethics/>

- No

13. Rate the paper on a scale of 1-6, 6 being the strongest (6-4: accept; 3-1: reject). Please use the entire range of the distribution. Spreading the score helps create a distribution for decision-making.

- 4. Weak Accept — marginally above the acceptance threshold, but would not mind if rejected, dependent on rebuttal

14. Please justify your recommendation. What were the major factors that led you to your overall score for this paper?

- Please refer to my previous comments.

16. In view of your answers above and your overall experience, how would you rate your confidence in your review?

- Somewhat confident (2)

Reviewer #4

Questions

3. Please categorize the relevance of the paper (you may choose more than one). Note the different assessment criteria for the different paper categories as outlined in the Reviewer Guidelines:

<https://conferences.miccai.org/2026/en/REVIEWER-GUIDELINES.html>

- MIC

4. How would you describe the paper?

- Both

5. Please describe the main contribution of the paper.

- The paper introduces LongMedBench, a benchmark for evaluating medical agents on long-horizon clinical decision-making using longitudinal EHR data from MIMIC-IV. It constructs dense multi-visit patient trajectories and proposes a hierarchical evaluation framework including factual QA, temporal reasoning, and long-horizon decision tasks. The benchmark aims to assess how models utilize long-term clinical history for reasoning and planning.

6. Please list the major strengths of the paper. For example, you could highlight a novel formulation, an original way to use data, a demonstration of clinical feasibility, a novel application, a particularly strong evaluation, or anything else that is a strong aspect of this work. Please provide details, for instance, if a method is novel, explain what aspect is novel and why this is interesting.

- Strengths
 1. The paper clearly identifies the limitation of existing medical agent benchmarks focusing on short-context QA and tool use, and motivates long-horizon clinical reasoning, which is meaningful and important.
 2. It constructs a longitudinal EHR-based benchmark from MIMIC-IV with dense multi-visit trajectories, enabling evaluation beyond single encounters.
 3. The hierarchical task design (factual QA → temporal reasoning → long-horizon decision) provides a structured way to evaluate different reasoning levels.

7. Please list the major weaknesses of the paper. Please provide details: for instance, if you state that a formulation, way of using data, demonstration of clinical feasibility, or application is not novel, then you must provide specific references to prior work.

- Weaknesses

Benchmark details

1. The data filtering strategy keeps only abnormal lab results and patients with ≥ 15 hospitalizations. This may introduce strong observation bias, overrepresenting high-acuity patients and removing normal clinical signals.
2. In Sec. 2.2 (Long-Horizon Decision Making), the ground truth is sampled from historical actions. However, clinical decision-making often allows multiple valid next steps. The current formulation still assumes a single correct answer, even with the 24h time-decay mechanism. It is unclear how clinical correctness is ensured under this setup.
3. There is an inconsistency in the reported dataset size. The abstract mentions 335 patients, while Fig. 1 and Sec. 2.1 report 355 patients.

Experiment

1. Temporal reasoning performance is strong with explicit timestamps but drops significantly without them. Does this indicate that models rely heavily on explicit temporal markers rather than true implicit temporal reasoning, especially in visit-level and joint sorting tasks?
2. Increasing history length or introducing RAG and Mem0 brings limited improvement in long-horizon decision tasks. Does this suggest that the benchmark does not strongly depend on long-term memory as intended, and that performance is mainly driven by immediate context?
3. Factual QA scoring relies on LLM-based matching and keyword coverage. How is the reliability of this evaluation validated, for example agreement with human annotations or robustness to paraphrasing?

Presentation

1. There are typos in Fig.2, "Cheif Complaint" should be "Chief Complaint".
2. Figures are hard to read. The font size is too small, especially in Fig. 2, and the figure is overloaded with information.
3. Some figure content is redundant with the method section, and repeating equations in the figures is unnecessary and affects readability.

8. Please rate the clarity and organization of the paper.

- Good

9. Please comment on the reproducibility of the paper. Please be aware that providing code and data is a plus, but not a requirement for acceptance.

- The submission does not mention open access to source code or data, but provides a clear and detailed description of the algorithm to ensure reproducibility.

10. Code of Ethics Check. Based on your review and your understanding of the MICCAI Scientific Code of Ethics, do you believe this submission may involve a potential ethics concern or violation? Reviewers are expected to familiarize themselves with the MICCAI Scientific Code of Ethics before completing this assessment: <https://miccai.org/index.php/about-miccai/policies/scientific-code-of-ethics/>

- No

13. Rate the paper on a scale of 1-6, 6 being the strongest (6-4: accept; 3-1: reject). Please use the entire range of the distribution. Spreading the score helps create a distribution for decision-making.

- 4. Weak Accept — marginally above the acceptance threshold, but would not mind if rejected, dependent on rebuttal

14. Please justify your recommendation. What were the major factors that led you to your overall score for this paper?

- The paper addresses an important problem and proposes a benchmark for long-horizon clinical reasoning with a clear task design. Overall, it is important, but there are still some concerns. The data filtering may introduce bias, and the decision tasks rely on historical actions as ground truth, which may not reflect real clinical settings.

16. In view of your answers above and your overall experience, how would you rate your confidence in your review?

- Somewhat confident (2)

Meta-Reviewer #1

Meta-Review Questions

7. Please justify your decision. In case you deviate from the reviewers' recommendations, explain in detail the reasons why. In case of an invitation for rebuttal, clarify which points are important to address in the rebuttal.

- The reviewers appreciated the motivation/importance of the study, the task formulation (which is very timely), and the findings discovered by this curated dataset. As the meta-reviewer of this paper I personally like this paper, but I still wanna invite the authors for a rebuttal to address some potential small concerns raised by reviewers:
 1. The potentially introduced data bias which keeps only abnormal lab results and patients with ≥ 15 hospitalizations
 2. Experiments questions raised by R4 (they are very meaningful questions)
 3. The differentiation to other related benchmarks (R2)
 4. Some presentation problems and typos raised by R1 and R4

Comment by Codex

我建议这份 rebuttal 不要“逐条解释所有小问题”，而是紧贴 Meta-Reviewer 点名的 4 类问题写。目标很明确：稳住两个 Weak Accept，让 Reviewer #1 觉得主要疑虑已经被认真处理，至少不继续强烈反对。

最重要的主线

1. 先主动降 novelty 口径

不要硬说“first / new direction”。R2 的点是合理的：MedAgentBench、AgentClinic、TIMER、ER-Reason 都已经在往 medical agent / temporal EHR reasoning 方向走。更好的说法是：

We agree that LongMedBench should be positioned as a temporally focused complement to existing medical-agent benchmarks, rather than an entirely new direction.

然后马上区分你们的贡献：LongMedBench 的重点不是“有 medical agent”，而是 **dense multi-visit EHR trajectories + leakage-controlled memory setting + explicit/implicit temporal reasoning + memory architecture stress test**。

2. 把 long-horizon decision-making 改口为“retrospective action-consistency proxy”

这是 rebuttal 里最关键的防守点。现在正文里“optimal next medical action / true long-horizon reasoning”的口气太满，R2/R4 都抓住了：历史 EHR action 不等于唯一正确临床决策。

建议承认：

This task is not intended to certify optimal clinical treatment. It evaluates whether an agent can recover clinically plausible next-step actions consistent with the observed care trajectory under the available historical context.

这样你们不是在硬辩“EHR action 就是 gold standard”，而是在重新定义 benchmark 的评价对象：**trajectory continuation / decision consistency**。同时强调 24h time-decay scoring 是为了缓解 multiple valid actions 的问题，而不是彻底解决它。

3. 数据 bias 要承认，但解释为 stress-test cohort

R4 和 Meta 都点名了 abnormal labs + ≥ 15 hospitalizations。不要辩成“没有 bias”。建议说：

We agree this filtering creates a high-acuity, high-utilization cohort and should not be interpreted as population-representative.

然后解释为什么这么做：你们是为了构造足够长、事件足够密的 longitudinal trajectories，专门 stress-test memory 和 temporal reasoning。也可以补一句：所有模型在同一 cohort 上比较，所以不影响相对评估，但会影响外推性，你们会在 limitation 里明确。

4. 把“历史越多帮助有限”解释成 **negative finding**，而不是 **benchmark 失败**

R4 问得很尖锐：如果加 memory/RAG/Mem0 都没提升，是否说明 benchmark 不依赖 long-term memory?

建议回答：

We interpret this as a negative result about current agents, not as evidence that history is irrelevant. The benchmark exposes that existing memory mechanisms retrieve facts better than they integrate longitudinal evidence for decisions.

但这里要小心，别说得太绝对。你们自己的 Table 结果确实显示 immediate context 很强，所以最好同时降调：

We will revise the wording to avoid implying that the decision task fully isolates long-horizon reasoning.

5. 直接认错两个 **consistency** 问题

这里不要解释太多，直接说会修：

- [MICCAI2026-main conference paper template.tex]
(/Users/cometp/Documents/research/LongMedBench/MICCAI2026-main conference paper template.tex:81) abstract 写 335 patients;
- [MICCAI2026-main conference paper template.tex]
(/Users/cometp/Documents/research/LongMedBench/MICCAI2026-main conference paper template.tex:188) method 写 355 patients。

按 6,999 visits / 19.72 visits per patient 看，应该统一成 **355**。另外 [line 519]

(/Users/cometp/Documents/research/LongMedBench/MICCAI2026-main conference paper template.tex:519) 说 all models 128K，但 Table 里 GPT-5-mini 是 400K，也要认作 wording error。

我会建议的 **rebuttal** 结构

第一段：感谢 reviewers，并说我们会 focus on four concerns.

第二段：Novelty / related benchmarks

承认已有工作，强调差异，不要硬抢 “first”。

第三段：Data filtering bias

承认 high-acuity bias，解释 stress-test 目的，承诺 limitation 中明确。

第四段：Decision task ground truth + long memory

承认不是 optimal clinical decision，定义为 observed trajectory consistency；解释 time-decay；把 memory 结果解释为 current agents 的 failure mode。

第五段：Temporal reasoning / scoring / reproducibility

说 explicit-vs-implicit gap 正是 benchmark intended to reveal；factual QA labels mostly come from structured event fields，LLM/keyword matching只用于容忍 free-text paraphrase，camera-ready 会补 scoring prompt/rubric/pipeline detail。不要声称做了 human agreement，除非原稿已经有。

第六段：Presentation

一两句话：修 typo、figure readability、Mem0 explanation、medical event definition、patient notation、bold text。

一句话版本的策略

你们应该把 rebuttal 的姿态从“我们完全没问题”改成：

The reviewers are right that the scope and claims need sharpening; the benchmark’s value is not claiming optimal clinical decision-making, but exposing how current agents fail to use dense longitudinal EHR histories beyond explicit retrieval and timestamp cues.

这比逐条防守更容易打动 AC。

我也核对了几个相关 benchmark 的定位，方便你们写 related-work 区分：MedAgentBench 更偏 EHR agent task environment，AgentClinic 偏 simulated clinical interaction，TIMER 是 longitudinal EHR temporal instruction/evaluation，ER-Reason 偏 ER workflow reasoning with clinician rationales。对应来源：MedAgentBench / AgentClinic / TIMER / ER-Reason：

- <https://arxiv.org/abs/2501.14654>
- <https://arxiv.org/abs/2405.07960>
- <https://www.nature.com/articles/s41746-025-01965-9>
- <https://arxiv.org/abs/2505.22919>

Comment by DS

审稿结果总览

审稿人	分数	决定	关键词
R1	3/6 - Weak Reject	取决于 rebuttal	写作差、贡献增量小、图不清楚、不可复现
R2	4/6 - Weak Accept	取决于 rebuttal	问题重要、过度声称novelty、与5个benchmark区分不够

R4 4/6 - Weak Accept 取决于 rebuttal 设计合理但数据偏差、ground truth有问题

Meta 邀请 rebuttal — 个人喜欢这篇、要求4点回应

需要回应 (Rebuttal) 的 4 大核心问题 (Meta-Reviewer 指定)

1. 数据偏差 (R4 指出)

只保留异常 lab 结果 + 只选 ≥ 15 次住院的病人 $\rightarrow$ 过度代表高危病人, 丢失正常临床信号

2. 实验相关问题 (R4 提出, Meta 特别强调"非常有意义")

- 时序推理去掉 explicit timestamp 后大幅下降 $\rightarrow$ 模型依赖显式时间标记而非真正的 implicit temporal reasoning?
- RAG/Mem0/增加历史对 long-horizon decision 帮助有限 $\rightarrow$ 是不是说明 benchmark 实际上不强依赖长期记忆, 主要由 immediate context 驱动?
- LLM-based evaluation 的可靠性 (factual QA) 你如何验证?

3. 与相关工作区分 (R2 指出)

引用 5 个已有 benchmark (MedAgentBench, AgentClinic, TIMER, ER-Reason, EHRSQL), 认为 novelty 在 task packaging 而非整体 idea

long-horizon decision task 预测下一步 EHR action 更像是 trajectory continuation 而非 gold-standard clinical decision

4. 写作/图表/typos (R1 + R4)

R1 列了很多小问题: semicolon 不再大写、过多 bold、页边距溢出、"a action" $\rightarrow$ "an action"、"is" $\rightarrow$ "are"、"Cheif Complaint" $\rightarrow$ "Chief Complaint"、图2字体太小/信息过载、Mem0 不是引入的、ref 大写不正确……R4 补充了 Figure 2 typos 和格式冗余。

我的分析

这篇 rebuttal 的关键筹码:

- 两个 Weak Accept + 一个 Weak Reject, 不是绝望局。Meta 说了 "I personally like this paper", 天平在你这边。
- R1 的 reject 理由基本是 writing/presentation——最容易在 rebuttal 里回应的东西 (已经在改了、会改完)。
- R2/R4 的技术疑问同一篇论文能回应: 数据 bias、novelty 区分、decision task 的 ground truth 本质、implicit temporal reasoning 是否真实。这几个问题是核心防线。

R1 和 R4 都提到的 335 vs 355 patients 不一致 是硬伤, rebuttal 里必须正面承认并解释。

TODO

画图问题：信息量太大，但是空间有限，如何权衡

修改&回应 R1: Moreover, the contribution is incremental.

Its substantive distinction is the combination of: (i) dense real EHR trajectories (355 patients, 19.72 visits/patient); (ii) controlled note/event/context memory granularities; and (iii) a progressive evaluation that separates factual lookup, implicit temporal reconstruction, and next-step decision tasks. This setting targets cross-visit temporal integration, which prior benchmarks do not systematically measure.

可以看intro中修改的部分。

回应 R1: the work is not reproducible.

We thank the reviewer for raising the reproducibility concern. We agree that reproducibility is essential for benchmark work. Since LongMedBench is built on MIMIC-IV, a credentialed-access clinical database, we are not allowed to publicly redistribute the raw EHR records or processed patient-level samples, even after preprocessing. This restriction follows the data-use requirements of MIMIC-IV rather than a limitation of our method.

To support reproducibility within the permitted data-use boundary, we will release the complete benchmark construction pipeline, including cohort selection criteria, preprocessing scripts, event-stream construction code, prompt templates, evaluation scripts, model configurations, and random seeds. We have already implemented this pipeline in a Git repository, which is currently private due to the review-stage/anonymity requirement. Once the paper can be publicly disclosed, we will make the repository public. Any researcher with authorized MIMIC-IV access will be able to regenerate the benchmark samples and reproduce our experiments from the released code.

修改&回应 R2: overclaim novelty.

我们的novelty 主要在于 以上提到的 benchmark 没有针对长程医疗事件做专门特化，而这个在医学实际中很常见

新增两行（建议插在 EHRSQL 之前或之后）：

Benchmark	Long.	Ctx.	EHR	Dec.	Temp.	Focus
TIMER	✓	×	✓	×	✓	Temporal Relation QA
ER-Reason	×	×	✓	✓	×	Single-ED Workflow Reasoning
LongMedBench	✓	✓	✓	✓	✓	Long-horizon Reasoning

TIMER: Temporal Instruction Modeling and Evaluation for Longitudinal Clinical Records

ER-Reason: A Benchmark for Multi-Stage Clinical Reasoning in the Emergency Room

解释：

- TIMER: 有 Longitudinal（多时间点QA）和 Temporal Sensitivity（时间边界、趋势检测、时序排序），但没有 Multi-turn Context 和 Decision-making。它的核心是静态的时间推理QA，而非交互式agent决策。
- ER-Reason: 有 EHR（真实急诊数据）和 Decision-making（分诊→诊疗→出院决策），但它的纵向维度仅限于单次急诊 episode 内的流程，不是跨多年的多住院轨迹；没有显式的 temporal reasoning 评估。

这样修改后，Table 1 直观展示 LongMedBench 是唯一同时覆盖全部五个维度的 benchmark。

修改&回应 R2: Third, the paper's own ablations show that adding more historical memory gives limited benefit which weakens the claim that the benchmark already isolates a strong long-horizon reasoning / **回应 R4:** Increasing history length or introducing RAG and Mem0 brings limited improvement in long-horizon decision tasks. Does this suggest that the benchmark does not strongly depend on long-term memory as intended, and that performance is mainly driven by immediate context?

这不是benchmark设计失败。恰恰相反——factual QA中记忆增强有效（Mem0 Lab-T: 0.99），而 decision task中无效。同一个benchmark下，记忆帮助检索但不帮助决策，这正是LongMedBench的核心诊断价值：它成功分离了"信息可用"和"信息整合"两个能力。如果RAG解决了所有任务，benchmark才是真正没有区分度。

修改&回应 R2: predicting the next recorded EHR action is closer to trajectory continuation than gold-standard decision evaluation. / **回应 R4:** In Sec. 2.2 (Long-Horizon Decision Making), the ground truth is sampled from historical actions. However, clinical decision-making often allows multiple valid next steps. The current formulation still assumes a single correct answer, even with the 24h time-decay mechanism. It is unclear how clinical correctness is ensured under this setup.:

代码块

- 1 防御层1：重新定义评估目标（回应R1）
- 2 > 该任务评估的不是"模型能否做出最优临床决策"，而是"模型能否利用长篇历史信息区分临床上与特定患者一致/不一致的动作"。

3 - **类比**：阅读理解任务不声称"模型给出的答案是最优解释"，而是"基于文本，哪个选项是正确答案"。真值来自文本，不是来自外部真理。LongMedBench同理——真值来自EHR轨迹，评估目标是模型**理解并利用历史上下文**的能力。

4 - **实证支持**：论文实验结果自身证明这不是trivial的轨迹复制：

5 - 如果只是"记忆轨迹"，为什么RAG和Mem0几乎不提升性能？

6 - 为什么模型排名在factual QA和decision-making上不同？

7 - 这暗示任务确实测到了非检索性的能力。

8 ### 防御层2：多选题 + 临床Plausible Distractor机制（回应R1+R2）

9 > 我们**不是** open-ended 生成评估。每个问题是**多选题**，distractor由LLM生成，要求"临床上看似合理，但基于该患者特定历史上下文是错误/次优的"。

10 这意味着：

11 - 正确答案**不需要**是"唯一最优的临床动作"——它只需要是"比distractors更符合该患者历史上下文的动作"

12 - 评估测量的是**相对判别能力**（given history, can you tell which action actually happened?），而非绝对正确性

13 - 类比：这不是考"医生应该开什么药"，而是考"这个病人的病历里下一个药实际是什么"——后者是一个信息理解题，不是决策最优性题

14 - 如果某一时刻真有多个完全等价的合理动作，LLM生成的distractor无法与正确答案区分，那么该样本的难度对模型是合理的——因为需要精细的历史信息来做出区分

15 ### 防御层3：24h时间衰减机制更细致的辩护（回应R2）

16 > ReviewR2认为"即使有24h衰减，仍然假设单一正确答案"。但24h衰减在数学上**明确承认了多有效动作**。

17 ...

18
$$S = \max_{i \in E_{24h}} (e^{-\lambda(t_i - t)} \times \text{Score}(a_i, \hat{a}))$$

19 ...

20 - 如果模型预测的动作在24h内**任何位置**出现，都能获得部分credit

21 - max算子意味着只要匹配到窗口内的任意一个事件即可得分，而非强制匹配唯一的GT事件

22 - 时间越近权重越高——反映临床实际：越临近的动作越相关

23 - **这不是"单一正确答案"，而是"加权多正确答案"——模型不必精确命中GT事件，只需命中时间窗口内的任何临床相关事件

24 **可用例子说明**（适合放在rebuttal中）：

25 > 假设GT事件是"12:00开化验单A"，但模型预测了"12:30开化验单B"。如果12:30确实有化验单B的记录，且权重仍然较高，模型仍可获得部分分数。这明确不是单一正确答案假设。

26 ### 防御层4：临床正确性的间接保证（回应R2）

27 虽然没有直接的expert validation，但通过以下机制间接保证：

28 | 机制 | 作用 |

29 |-----|-----|

30 | MIMIC-IV真实数据 | 由真实医生在实际约束下做出的决策，不是合成/随机数据 |

31 | ≥15次住院筛选 | 密集轨迹意味着后续住院与前序临床决策有**实际因果关系**，而非random noise |

32 | LLM distractor prompt约束 | "在临床上合理但在该具体场景下错误"，引入了临床合理性检查 |

33 | 时间衰减窗口 | 承认临床等价动作的存在，不要求精确匹配 |

34 ### 防御层5：这是领域共性限制（回应R1）

35 | Benchmark | 如何处理decision ground truth? | 同样面临此问题? |

36 |-----|-----|-----|

37 | MedAgentBench | 使用EHR中记录的实际医嘱作为真值 | ✓ |

38 | ER-Reason | 使用实际临床决策（诊断、分诊、出院）作为真值 | ✓ |
39 | AgentClinic | 使用预设的"正确"交互脚本 | ✓ |
40 | DiagBench | 使用EHR诊断轨迹作为真值 | ✓ |
41 | **LongMedBench** | 使用EHR轨迹 + 时间衰减 + LLM distractor | ✓ 但有额外缓解机制 |
42 **结论**：任何基于回顾性EHR数据的benchmark都面临此问题。LongMedBench的优势在于通过三个
额外机制做了更多缓解，且**不声称评估决策最优性**。

43 ---

44 ## 三、Rebuttal 建议框架（三步结构）

45 ### Step 1 – 承认限制（1句话，不回避）

46 > We agree that recorded EHR actions are not equivalent to prospectively
validated gold-standard decisions, and that multiple clinically valid actions
can exist at any point. We do not claim that the task evaluates decision
optimality.

47 ### Step 2 – 三重防御（核心段落）

48 > Instead, the task is designed to measure **context-discriminative ability**:
whether the agent can identify which action is most consistent with a
patient's longitudinal history. Three design choices mitigate the trajectory-
continuation concern:

49 >

50 > ** (a) Multiple-choice format with clinically plausible distractors. ** Each
question presents the ground-truth action alongside LLM-generated distractors
that are clinically reasonable in general but inconsistent with the patient's
specific trajectory. This transforms the evaluation from "which action is
globally optimal" to "which action is most consistent with this patient's
historical context." The model must leverage long-horizon memory to make fine-
grained discriminations.

51 >

52 > ** (b) 24h time-decay scoring. ** The scoring function $(S = \max_i \{ \text{in} \mathcal{E}_{24h} \} e^{-\lambda(t_i - t)} \cdot \text{Score}(a_i, \hat{a}))$
explicitly rewards any model-predicted action occurring within 24 hours of the
target timestamp, with decaying weight. This acknowledges that temporally
proximate clinical actions can be equally valid—a model is not penalized for
predicting a nearby alternative rather than the exact recorded event.

53 >

54 > ** (c) Empirical validation of task design. ** If the task merely measured
trajectory memorization, memory augmentation (RAG, Mem0) would be expected to
significantly boost performance. Our finding that it does not (Table 5:
marginal differences between RAG/Mem0 and baseline) suggests the task probes
deeper clinical reasoning that is not trivially resolved by retrieving more
history. This is consistent with the task measuring context-grounded
discrimination rather than simple pattern matching.

55 ### Step 3 – 定位为future work, 展现诚意

56 > We acknowledge this as an inherent limitation of retrospective EHR
benchmarks and have added an explicit discussion in the Limitations paragraph
of the revision. Expert-annotated evaluation—where multiple clinically valid
actions are pre-identified per state, or where predictions are cross-

referenced against clinical practice guidelines—represents important future work that would complement the current discriminative framework.

回应 R4: The data filtering strategy keeps only abnormal lab results and patients with ≥ 15 hospitalizations This may introduce strong observation bias, overrepresenting high-acuity patients and removing normal clinical signals.

R4 raises concerns about filtering abnormal labs and patients with ≥ 15 hospitalizations. Our strategy is driven by task feasibility: MIMIC-IV event streams are sparse for most patients, and short trajectories cannot support longitudinal reasoning evaluation. The ≥ 15 criterion yields sufficient density for multi-step reasoning. Abnormal-only labs preserve clinically salient signals while avoiding large context inflation. We acknowledge this creates a high-acuity subpopulation and will discuss the scope limitation and generalizability implications.

关键引用

Hager et al. (2024), Nature Medicine — 最直接的先例:

"filtering the laboratory results and removing all normal test results generally improved performance"

"This allows the model to focus more on abnormal, pathological signals."

"presenting the model with only abnormal laboratory results generally improved diagnostic performance."

该文在 MIMIC-IV 上用 only_abnormal_labs 标志实现了完全相同的策略，并在 Nature Medicine 发表。这是我们最强的防守武器。

Roy et al. (2018, AMIA) 和 Lin et al. (2019, JAMA Network Open) 提供了"正常结果信息密度低"的概念基础，但不必引入 rebuttal——一篇 Nature Medicine 先例足以。

论文修改 (Sec 2.1.1, line 190 末尾加一句)

% 当前 (line 190)

...rigorously evaluates the agent's ability in cross-session memorization and temporal reasoning.

% 后面加:

We retain only abnormal labs following prior work showing that normal results reduce signal density and degrade LLM diagnostic performance \cite{hager2024evaluationlimitations}; we

acknowledge this yields a high-acuity cohort and do not claim generalizability to lower-utilization populations.

需在 ref.bib 中加入：

```
@article{hager2024evaluationlimitations,  
  title={{Evaluation and mitigation of the limitations of large language models in clinical decision-making}},  
  author={Hager, Paul and Jungmann, Friederike and Holland, Robbie and Bhagat, Kunal and Hubrecht, Inga and Knauer, Manuel and Vielhauer, Jakob and Makowski, Marcus and Braren, Rickmer and Kaissis, Georgios and Rueckert, Daniel},  
  journal={Nature Medicine},  
  volume={30},  
  pages={2613--2622},  
  year={2024},  
  doi={10.1038/s41591-024-03097-1}  
}
```

Rebuttal 回应

We agree this creates a high-acuity cohort, but argue the filtering is well-motivated:

- (1) Abnormal-only lab filtering is established practice. Hager et al. (2024, Nature Medicine) demonstrated that retaining only abnormal laboratory results improves LLM diagnostic performance on MIMIC-IV, noting that normal results "reduce the amount of healthy information, which tends to confuse the tested models" and "allow the model to focus more on abnormal, pathological signals." Our approach follows this same evidence-based rationale.
- (2) Necessity for longitudinal evaluation. The ≥ 15 hospitalization criterion is required for task feasibility: MIMIC-IV event streams are sparse for most patients, and trajectories with <15 visits cannot support the multi-step reasoning our benchmark evaluates. This is analogous to how long-context benchmarks like RULER require minimum sequence lengths.
- (3) Targeted scope. LongMedBench is designed to stress-test long-horizon reasoning where clinical failures are most consequential—complex, high-utilization patients. We have added an explicit scope statement in the revision acknowledging that findings do not directly generalize to routine or low-acuity populations.强调

强调如果不移除lab会有过多的冗余信息，因为大多数的指标都是正常的（上面的工作证明了）。所以在后续的benchmark中为了保证公平性我们移除了对lab事件和参数的预测。未来的benchmark会弥补这一点。

回应 R4: temporal reasoning performance is strong with explicit timestamps but drops significantly without them. Does this indicate that models rely heavily on explicit temporal markers rather than true implicit temporal reasoning, especially in visit-level and joint sorting tasks

审稿人的质疑恰好是论文自己报告的发现（line 414-417）：

"top models achieve near-perfect performance (gpt-5-mini: 0.925) ... Moving to visit sorting, where timestamps are absent ... performance drops sharply"

论文不需要改。rebuttal中一句话化解：We agree with the reviewer—this is precisely the diagnostic finding LongMedBench enables. The paired design of visit_cloze (timestamps provided) and visit_sorting (timestamps removed) isolates implicit temporal reasoning as the bottleneck. The sharp drop from 0.925 to 0.423 demonstrates that current LLMs rely on explicit temporal markers rather than internalizing clinical progression cues from narrative context.

回应 R4: Factual QA scoring relies on LLM-based matching and keyword coverage. How is the reliability of this evaluation validated, for example agreement with human annotations or robustness to paraphrasing?

当前论文没有说明评分怎么做的。R4问得很具体：和人工标注的一致性？对改写/同义词的鲁棒性？

我们要求模型返回和gt完全一致的内容，必须完全照搬，不得包含缩写/扩写/改写。对于不同 Action/parameter类型，各自设计了对应的few-shot提示，用以规范Agent的返回格式。例如，GT要求返回检查名称“Arterial Blood Gas”，如果模型返回“ABG”或者“Arterial Blood Gas Analysis”，这会被视为0分的错误答案。这样的设计参考了RULER（Hsieh et al., 2024）以及 NeedleBench（Li et al, 2025）等通用领域框架，确保模型具有准确的检索能力——如果不能复现原始数据，那么检索的能力就是有缺陷的。重点放在模型能力的评估。

回应&修改 Meta Reviewer —— rebuttal回应的骨架可以按照这个写：

1. The potentially introduced data bias which keeps only abnormal lab results and patients with ≥ 15 hospitalizations
2. Experiments questions raised by R4 (they are very meaningful questions)

3. The differentiation to other related benchmarks (R2)
4. Some presentation problems and typos raised by R1 and R4

语法问题/大小写相关：

最后grammarly扫一遍，已经修好了语法问题。

Rebuttal Draft

代码块

- 1 We thank the reviewers and the meta-reviewer for their careful feedback. We
2 address the four concerns below and will revise the paper accordingly.
- 3 1. Data Bias (abnormal labs, ≥ 15 visits) (R4, Meta Q1)
- 4
- 5 (a) Lab filtering follows MIMIC-CDM (Hager et al., 2024) from Nature Medicine,
6 which showed that normal ranges act as distractors in longitudinal reasoning.
7 We agree this creates a high-acuity cohort and will add a scope statement in
8 Sec. 2.1.1. However, this makes the evaluation more clinically realistic. Like
9 real-world clinicians, the agent should reason from salient signals, not
10 passively absorb all measurements.
- 11 (b) ≥ 15 visits balances sufficient length for constructing long-horizon tasks
12 and ensuring an adequate patient count. At 20 visits, only ~ 100 patients
13 remain; at 10, visit-level sorting/ranking tasks become infeasible. At 15, we
14 retain 355 patients, comparable to existing benchmarks: 276 for MedAlign
15 (Fleming et al., 2024), and 511 for DiReCT (Wang et al., 2024). We acknowledge
this scope limitation and will discuss this trade-off in the revision.
2. Experiment Questions (R4, R2, Meta Q2)
- We respectfully argue that the majority of R4's comments on the experiments
are empirical findings, not design flaws. They reveal model limitations in
implicit temporal reasoning, memory mechanisms, and history integration.
- (a) Implicit temporal reasoning (R4). The τ drop without timestamps reveals
that current LLMs have not internalized temporal structure from clinical
narratives, a deficit invisible to prior temporal QA benchmarks. Our paired
design intentionally isolates this bottleneck.
- (b) Marginal benefit from RAG/Mem0/history (R2, R4). The negative result
indicates that models can retrieve but cannot integrate cross-visit evidence
into decisions. Factual QA degrades with history; decision-making stays flat

regardless of memory. Together, they show retrieval is history-dependent while reasoning is not.

(c) Evaluation reliability (R4). For factual QA, to measure precise retrieval, models are prompted with few-shot examples to return the exact ground-truth values without paraphrasing. Answers are evaluated by exact keyword matching following works in general domain like RULER (Hsieh et al., 2024) from COLM 2024. We will add the detail to the task settings.

(d) Decision making (R2). We agree that the next recorded EHR action is not a gold-standard decision. This should be interpreted as whether an agent can combine immediate context with long-horizon history to recover a trajectory-consistent, clinically plausible near-future action. The recorded action is thus an observational reference. Our 24h decay provides tolerance for temporally adjacent actions, but we will explicitly state that it only partially addresses the existence of multiple valid next steps.

3. Contribution & novelty (R1, R2; Meta Q3)

We agree with the partial overlap with recent work. However, our claim is that, LongMedBench is not the first realistic longitudinal medical benchmark in general, but to our knowledge, the first benchmark that jointly evaluates agent-oriented clinical decisions on cross-visit real-world EHR trajectories under controlled note, event, and context memory settings. This setting is distinct from MedAgentBench's (Jiang et al., 2025) EHR tool-use tasks, AgentClinic's (Schmidgall et al., 2024) single-encounter diagnostic simulations, TIMER's (Cui et al., 2025) temporally grounded instruction evaluation, and ER-Reason's (Mehandru et al., 2025) ER-workflow belief-updating benchmark. We will fix the paper's novelty claim to this combination.

4. Presentation (R1, R4, Meta Q4)

Presentation issues will be corrected: the patient count unified to 355, typos fixed, the Mem0 explanation added, and Fig. 2 redesigned with larger fonts and adequate information. We will release dataset construction scripts and evaluation code upon acceptance.

Confidential Comments to Area Chairs and Program chairs (4000 chars max)

代码块

1 We appreciate the reviewers' feedback and believe the major technical concerns can be largely resolved in the rebuttal.

2

3 We respectfully note that the reviews differ substantially in technical
engagement. R2 & R4 provided detailed comments on benchmark positioning,
longitudinal decision-making, and interpretation of the lab results, which
helped clarify several aspects of the paper. These concerns are primarily
about framing and interpretation rather than fundamental methodological flaws.

4

5 In contrast, R1's assessment focuses predominantly on presentation issues. The
main technical criticism ("incremental contribution") was not accompanied by
specific prior work or technical comparison, unlike R2's concrete discussion
of related benchmarks. We therefore respectfully hope the AC can consider the
depth of technical analysis across the reviews when weighing the overall
assessment.

6

7 We would also like to clarify one central point underlying the ablation
discussion. The limited gains from additional memory and the large gap in
temporal reasoning are intended empirical findings of the benchmark rather
than unintended failures of the setup. The benchmark is specifically designed
to expose the gap between retrieval and true longitudinal reasoning under
realistic multi-visit EHR trajectories, and the rebuttal clarifies this
distinction explicitly. We hope the paper meets the acceptance threshold and
respectfully request consideration for acceptance.